

Multimodal PDAC Metastatic Biomarker Discovery

1. Research Objective

The core objective is to develop a computational framework that integrates **Histopathology Foundation Model embeddings** with **Spatial Transcriptomics (ST)** to identify specific morphological features in primary pancreatic tumors that predict liver metastasis.

Crucially, the goal is to "augment" tissue embeddings during training so that, at inference time, the model can predict metastatic potential directly from H&E images without requiring ST data.

2. Research Phases

Phase A: Knowledge Distillation & Embedding Augmentation

Objective: To enrich morphological features with biological ground truth, creating "augmented embeddings" that capture more than just visual shapes.

- **Foundation Model Feature Extraction:** Utilize state-of-the-art pathology foundation models to extract high-dimensional embeddings from H&E patches.
- **ST-Morphology Fusion:** Align these embeddings with the corresponding Spatial Transcriptomics spots.
- **The Augmentation Task:** Train a projection head or a transformer-based architecture that learns to represent the transcriptomic profile within the visual embedding space.
- **Key Outcome:** A model capable of **ST-free inference**—taking an H&E slide and generating an "augmented embedding" that inherently contains the predicted biological state of those cells.

Phase B: Identifying the "Metastatic Signature"

Objective: To distinguish between tumor cells that are "prone to metastasize" (specifically to the liver) and those that are not.

- **Primary vs. Metastatic Comparative Analysis:** * Analyze patches from the **Primary Tumor (PT)** and compare them against matched **Liver Metastasis (HM)** samples.
 1. Use the augmented embeddings from Phase A to identify which specific "spots" or "features" in the primary tumor most closely resemble the confirmed metastatic cells in the liver.

- **Feature Dissection:** Identify the visual/biological markers (e.g., specific cell-cell interactions or extracellular matrix patterns) that characterize the cells that "went to the liver."
- **Quantification:** Develop a scoring system to tell the clinician:
 1. Which specific spots in this primary tumor are already "metastasized" in their behavior.
 2. The likelihood of liver-specific metastasis based on current tumor morphology.

Phase C: Clinical Application & Discovery

Objective: To move from "Million Dollar" theory to clinical service.

- **Inference Pipeline:** Deploy the model on standard H&E whole slide images (WSI) where ST is unavailable.
- **Liver Metastasis Prediction:** Map the risk of liver metastasis across the entire primary tumor.
- **Biological Verification:** Identify which genes (from the ST used in Phase A) are the primary drivers of the morphological features identified in Phase B. This provides the "Why" behind the prediction.

Proposed Dataset:

To train a model that "sees" transcriptomics in H&E, we need the following "Gold Standard" pairs:

- **Matched Tissue Triads:**
 - **Primary Tumor (PT):** H&E WSI + corresponding ST data.
 - **Liver Metastasis (HM):** H&E WSI + corresponding ST data (from the same patient if possible).
 - **H&E Only Samples:** A larger cohort of clinical H&E slides (PT and HM) without ST for large-scale validation and inference.
 - Does we have samples from the **same patient** for both the primary and the liver metastasis? (This is the "Gold Mine" for this specific research).
- **Single-Cell RNA-seq (scRNA-seq) Reference:** If your ST is "spot-level" (like 10x Visium), you need a scRNA-seq atlas of PDAC to **deconvolve** those spots into specific cell types (e.g., distinguishing malignant epithelial cells from fibroblasts).
- **Clinical Metadata:** Specifically, the **site of metastasis** (confirming liver) and, if possible, the "time to metastasis" for patients where you only have the primary tumor.
- **Gene Panel Coverage:** Ensure your ST data includes known PDAC metastatic markers (e.g., *S100A4*, *POSTN*, *MMP* family, or EMT-related genes). If these genes have "zero counts" across your slides due to low sequencing depth, the model won't have a "target" to learn.
- **Mitochondrial Read %:** High mitochondrial DNA in spots usually indicates dying cells. These spots should be filtered out, as their morphology is "distorted" and will confuse the Foundation Model.

Proposed Methodology:

Phase 1: Data Engineering & Co-Registration

This is the most labor-intensive part. If the alignment is off by even a few microns, the model will learn noise.

- **Slide Pre-processing:** Perform **Macenko Normalization** on H&E slides to ensure staining consistency across different batches.
- **ST-H&E Alignment (Registration):** Use tools like **STalign** or **Squidpy** to map ST spot coordinates (x, y) to specific pixel coordinates on the WSI.
- **Transcriptomic Filtering:** * Select **Highly Variable Genes (HVGs)** (top 2,000–5,000) or focus on a **Metastatic Gene Panel** (e.g., EMT markers, PDAC-specific liver-tropism genes).
 - Apply log-normalization and scaling to the gene counts.

Phase 2: Feature Extraction (The Vision Branch)

Utilize the "frozen" intelligence of Pathology Foundation Models.

- **Patching:** Extract patches (e.g., $224 * 224$ pixels) centered exactly on the ST spots.
- **Embedding Extraction:** Pass patches through a model like **UNI** (from HMS) or **Virchow** (from Paige).
 - Input: H&E Patch
 - Output: 1024 or 1536 depending on the model).
- **Why Frozen?** We don't fine-tune the FM because it already knows "general pathology." We only train the "bridge" to the biology.

Phase 3: The Bridge (Embedding Augmentation)

This is where you solve Dr. Ashiq's question. You must create a **Latent Space** where vision and biology meet.

The Architecture: Contrastive Learning (CLIP-style)

- **Vision Projector (Pv):** A small MLP that maps z to a joint space S .
- **Biology Projector (Pb):** An MLP that maps the Gene Expression vector (g) to the same space S .
- **Training Objective:** Use **InfoNCE Loss** (Contrastive Loss).
 - Maximize similarity between a patch and its *actual* ST spot.
 - Minimize similarity between a patch and *other* ST spots.

The Alternative: Generative Distillation

- Train the model to **reconstruct** the gene expression vector from the visual embedding.
 - $\text{Loss} = \|\text{Decoder}(\text{Pv}(z)) - g\|^2$
- **Outcome:** The Visual Projector (Pv) now produces an "Augmented Embedding" that contains predicted transcriptomic information.

Phase 4: Metastatic Risk Mapping (The Discovery)

Once the "Bridge" is trained, you use the **Primary (PT) vs. Metastatic (HM)** samples to find the signature.

1. **Define the Reference:** Take the **Liver Metastasis (HM)** slides. Extract their augmented embeddings. This is your "True Metastatic Signature."
2. **The Comparison:** For a new **Primary Tumor (PT)** slide, calculate the cosine similarity between every PT patch embedding and the HM reference signature.
3. **Risk Scoring:** $\text{Similarity}(\text{PTpatch}, \text{HMAvg_signature}) = \text{Metastatic Potential}$
 - Generate a spatial **Heatmap** across the primary tumor showing exactly *where* the "liver-bound" cells are hiding.

Phase 5: Inference (The Clinical Service)

This is the final product you deliver to the clinic.

- **Input:** Standard H&E slide (No ST data).
- **Process:** 1. Patching → 2. Foundation Model → 3. **Trained Vision Projector (Pv)**.
- **Output:** A predicted transcriptomic map and a **Liver Metastasis Risk Score**.

Required Data (Before starting the project):

Patient ID	PT Sample	HM Sample	Image Name	Clinical Data	WSI H&E Image
Patient 1	Available	NA	IU_PDA_T1	Present	Present
Patient 2	Available	Available	IU_PDA_HM2	Present	Missing
			IU_PDA_HM2_2	Missing	Missing
			IU_PDA_T2	Missing	Missing
Patient 3	NA	Available	IU_PDA_HM3	Missing	Missing
			IU_PDA_T3	Present	Present
Patient 4	Available	Available	IU_PDA_HM4	Present	Missing
			IU_PDA_T4	Missing	Present
Patient 5	NA	Available	IU_PDA_HM5	Present	Missing
Patient 6	Available	Available	IU_PDA_HM6,	Missing	Missing
			IU_PDA_T6	Present	Missing
Patient 7	NA	NA	NA	NA	NA
Patient 8	Available	Available	IU_PDA_HM8,	Missing	Missing
			IU_PDA_T8	Present	Missing
Patient 9	Available	Available	IU_PDA_HM9,	Missing	Missing
			IU_PDA_T9	Present	Missing
Patient 10	Available	Available	IU_PDA_HM10,	Missing	Missing
			IU_PDA_T10	Present	Missing
Patient 11	Available	Available	IU_PDA_HM11	Missing	Present
			IU_PDA_T11	Present	Present
Patient 12	Available	Available	IU_PDA_HM12,	Missing	Missing
			IU_PDA_T12	Present	Missing
Patient 13	NA	Available	IU_PDA_HM13	Present	Present

Available = The sample was present in the dataset
NA = The sample did not exist in the dataset (would be great if we can have this)
Present = The data is present for the project
Missing = The data needs to be requested for the project

Since we need the PT and HM samples of the Patients:

- We currently have P1, P3 and P4 having only PT Images.
- We have P11 having both PT and HM
- We have P13 having only HM image

We need this:

- We need P2, P6, P8, P9, P10, P12 to get both of their PT and HM images.
- We need P3, P4 to get their HM images, while their PT are present

The total number of images to request is: 16

The below 7 .tiff files contain 4 image slides each, a mixture of known Zenodo samples and some entirely new samples. Also, 1 last .tif WSI is an unknown sample. The names are given below so we are not able to use these samples out of 30 samples shared.

1. 1_117.tiff
2. 2_119.tiff
3. 3_118.tiff
4. 120.tiff
5. V326.tiff
6. V327.tiff
7. V377.tiff
8. S21_23955_E7.tif

It is requested that all provided images be BIGTIFF-compatible, pyramidal, tiled TIFFs with JPEG compression, scanned at 40x and exported from the scanner's native software.

1. Clinical Data:

For reference, these are the data fields for clinical data:

Age	Gender	Ethnic	Location_in_pancreas
AJCC_stage	Tumor_Grade	R0_resection	Peripheral_Nerve_Inva

	_Histology		sion
Lymphovascular_Invasion	Nodes_positive	Neoadjuvant_Chemo	Treatment
Data_of_Surgery_start_point_in_survival_analysis	surgery_date	T_stage	N_stage
M_stage	nodes_pos	nodes_total	node_ratio

2. ST Data:

I have compiled a list of 62 metastasis marker genes through literature research, amongst which 6 genes were absent from the gene expression matrix:

- CDKN2A - Driver mutation
- TWIST1 - EMT / invasion
- MALAT1 - Epigenetic / ncRNA
- RACGAP1 - Prognostic signature
- RARRES3 - Prognostic signature
- IGF2BP3 - Signalling

The 6 missing genes are most probably caused by technical filtering while Seurat processing. Amongst these 6 genes: CDKN2A is one of the most driver mutations for liver metastasis, and MALAT1 is another abundantly expressed gene which was not present in gene expression matrix. Though their absence may affect the work, as genes are bundled up with other similar genes. But for similar grouped genes, one missing gene would not significantly trouble the model. If we can also get these, it would be great; otherwise, we can complete the experiment without these 6 genes.

Marker Gene Priority List for HM Samples (62 Genes):

Tier 1 — "Big Four" drivers (KRAS, TP53, SMAD4, CDKN2A): These are the most extensively validated. The accumulation of major driver alterations, including CDKN2A, KRAS, p53, and SMAD4, is inversely associated with disease-free and overall survival, and this altered gene combination is specifically linked to liver metastasis. SMAD4 loss in particular is the most strongly correlated single factor for metastatic recurrence.

Tier 2 — EMT/invasion genes: A global EMT expression signature identifies ~50 cancer cell clusters spanning the epithelial-mesenchymal continuum in both human and murine PDACs. Notably, total EMT suppression can *increase* liver metastasis via collective epithelial migration.

Tier 3 — Liver metastasis-specific: LITAF and SPRR1B show much higher expression in metastasis than primary PDAC samples, and high expression of these two genes is significantly

associated with poor prognosis in the TCGA dataset. These came from a validated 11-gene malignant metastatic cell signature derived from scRNA-seq of synchronously resected PT+LM tissue.

Tier 4 — TME/stromal markers: Specific subtypes of stromal and immune cells critical to the formation of the pro-tumour microenvironment in metastatic lesions include RGS5+ cancer-associated fibroblasts, CCL18+ lipid-associated macrophages, S100A8+ neutrophils, and FOXP3+ regulatory T cells.

For the CITED4 gene specifically — this is a brand new one from Theranostics 2025 that might be worth knowing, as it was validated functionally via siRNA knockdown in the same snRNA-seq + spatial transcriptomics framework.

3. Mitochondrial Level:

The dataset shows 0% for percent_mito for each PT and HM sample.